

CLASSIFICATION OF STATISTICAL MECHANICS

Every solid, liquid or gas in an assembly of an enormous number of microscopic particles, Likewise, radiation is an assembly of photons. Obviously, the actual motions or interactions of individual particles cannot be investigated. However, the macroscopic properties of such assemblies can be explained in terms of the statistical distribution of the individuals among different possible states and their most probable behaviour.

For example, from Maxwell distribution of speeds among the molecules of a gas we can calculate mean speed (which is related to the momentum carried by the molecules), mean square speed (which is related to the energy of the molecules) and so on. From the *average* quantities we calculate observable properties like pressure and temperature of the gas.

Usually we consider how a fixed amount of energy is distributed among the various identical particles of an assembly. Now there are three kinds of identical particles

- (i) Identical particles of any spin which are so much separated in the assembly that they can be distinguished from one another. The molecules of a gas are particles of this kind.
- (ii) Identical particles of zero or integral spin which cannot be distinguished from one another. These are called **Bose particles** (or bosons) and do not obey Pauli's exclusion principle. Photon, phonons and α -particle are of this kind.
- (iii) Identical particles of half integral spin which cannot be distinguished from one another. These are called **Fermi particles** (or fermions) and do obey Pauli's exclusion principle. Electrons, protons, neutrons are particles of this kind.

The first kind of the particles are the classical particles and obey the Maxwell-Boltzmann energy distribution law. The second and third kinds of particles are quantum particles and energy distribution laws for them can be derived by methods of quantum statistics (Bose-Einstein and Fermi-Dirac statistics) only.

4.4 EXCHANGE SYMMETRY OF WAVE FUNCTIONS

Let there be N particles described by the wave function $\psi(1, 2, 3, \dots, r, \dots, s, \dots, N)$ where 1 stands for all coordinates (position and spin) of particle 1 and so on. If the wave function is operated by an exchange operator $\hat{P}$ then

$$\hat{P}_{r,s} \psi(1, 2, 3, \dots, r, \dots, s, \dots, N) = P \psi(1, 2, 3, \dots, s, \dots, r, \dots, N) \quad \dots(27)$$

The effect of the operator $\hat{P}_{r,s}$ is to interchange the particles r and s i.e., formerly called particle ' r ' is now called ' s ' and vice versa. For two particles system, we write

$$\hat{P}_{12} \psi(1, 2) = P \psi(2, 1) \quad \dots(28)$$

Equation (27) is the eigenvalue equation of the operator $\hat{P}_{r,s}$ and P stands for the eigen-value of the operator $\hat{P}_{r,s}$.

There are *two* kinds of wave function ψ , depending on value of P – symmetric wave function and antisymmetric wave functions.

(i) Symmetric Wave Function (ψ_s)

A wave function is said to be symmetric if the interchange of any pair of particles among its arguments, leave the wave function unchanged.

$$\text{i.e.,} \quad P_{1,2} \psi_s(1, 2) = +\psi_s(2, 1) \quad (\text{here } P=1) \quad \dots(29)$$

(ii) Antisymmetric Wave Function (ψ_A)

The wave function is antisymmetric if the interchange of any pair of particles among its arguments, changes the sign of the wave function.

$$P_{1,2} \psi_A(1,2) = -\psi_A(2,1); \text{ here } P = -1 \quad \dots(30)$$

The exchange symmetry of the wave functions has close relation with the intrinsic angular momentum of the particle. The relationship is listed as follows :

- (a) The identical particles having an integral quantum number of their intrinsic spins are described by symmetric wave functions, such as

$$\psi_S(1,2) = +\psi(2,1) \quad \dots(31)$$

The type of particles described by symmetric wave functions are known as **BOSONS** as they obey Bose-Einstein statistics.

- (b) The particles which have half integral quantum number for their intrinsic spins are described by the antisymmetric wave functions, such as

$$\psi_A(1,2) = -\psi(2,1) \quad \dots(32)$$

The type of particles described by antisymmetric wave functions are called **FERMIONS** as they obey Fermi-Dirac statistics.

4.5.1 Applications of Bose-Einstein Statistics

(a) Planck's Radiation Law and Rayleigh Jeans Formula

Radiator or photon gas is considered as Bose-Einstein gas. Each photon has energy $h\nu$.

Let dn be the number of photons distributed in frequency interval ν to $(\nu + d\nu)$.

$$dn = n(\nu)d\nu = g(\nu)f(\nu)d\nu \quad \dots(43)$$

$$\therefore g(\nu)d\nu = \frac{8\pi h\nu^2}{c^3} d\nu \quad \dots(44)$$

$$\text{and } f(\nu) = \frac{1}{(e^{h\nu/k_B T} - 1)} \quad [\text{as } \alpha = 0 \text{ and } E = h\nu] \quad \dots(45)$$

Putting these values in Eq. (43), we get

$$dn = \frac{8\pi h\nu^2}{c^3} \cdot \frac{1}{(e^{h\nu/k_B T} - 1)} d\nu \quad \dots(46)$$

Let dE be the energy distributed in frequency interval ν and $(\nu + d\nu)$

$$dE = E d\nu = h\nu d\nu \quad \dots(47)$$

From Eqs. (46) and (47), we obtain

$$dE = E(\nu)d\nu = \frac{8\pi h\nu^3}{c^3} \cdot \frac{1}{(e^{h\nu/k_B T} - 1)} d\nu \quad \dots(48)$$

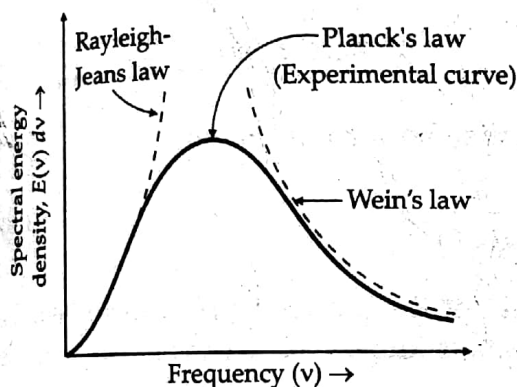
Equation (48) is known as **Planck's radiation formula** for the spectral energy density of black body radiation, which agrees with experimental curve (Fig. 4.4.)

For low frequency or high wavelength

$$(e^{h\nu/k_B T} - 1) = 1 + \frac{h\nu}{k_B T} - 1 = \frac{h\nu}{k_B T}$$

Equation (48) becomes as

$$dE = E(\nu)d\nu = \frac{8\pi\nu^2 kT}{c^3} d\nu \quad \dots(49)$$



Equation (49) is termed as *Rayleigh-Jeans law*. This law holds good for low frequencies and high wavelengths. Experimental agreement is shown in Fig. 4.4. For high frequency low frequency :

Equation (48) can be expressed in terms of wavelength of radiation as follows :

$$E_\lambda d\lambda = \frac{8\pi hc}{\lambda^5} \left(\frac{1}{e^{hc/(\lambda k_B T)} - 1} \right) d\lambda$$

or

$$E_\lambda = 8\pi hc \lambda^{-5} \left[\exp\left(\frac{hc}{\lambda k_B T} - 1\right) \right]^{-1}$$

Fig. 4.4 ν vs. $E(\nu)d\nu$ curve.

Let $\lambda_{\max}$ be the wavelength whose energy density is the greatest.

Then,
$$\frac{dE_\lambda}{d\lambda} = 0$$

Hence
$$\lambda_{\max} T = \frac{hc}{4.965 \text{ K}} = 2.898 \times 10^{-3} \text{ mK}$$

This is *Wien's displacement law*.

The peak in the black body spectrum shifts to progressively shorter wavelength (higher frequencies) as temperature is increased.

Spectral energy density decreases with frequency at high frequency. This is called as *Wein's law of energy distribution for black body radiation*.

Hence, Rayleigh-Jeans and Wein's laws are unable to explain the complete energy distribution for black body radiation, while Planck's law explains it completely, which is based on Bose-Einstein statistics.

4.8 COMPARISON OF M.B., B.E. AND F.D. STATISTICS

Maxwell-Boltzmann statistics (classical statistics)	Bose-Einstein statistics (Quantum statistics)	Fermi-Dirac statistics (Quantum statistics)
1. Apply to systems of distinguishable particles, such as gases.	Apply to systems of indistinguishable particles not obeying Pauli's exclusion principle, such as photons, phonons and liquid helium at low temperature	Apply to systems of indistinguishable particle obeying Pauli exclusion principle, such as free electrons in metal electrons in a star (white dwarf star)
2. The distribution law is $n_i = \frac{g_i}{e^{\alpha} e^{E_i/k_B T}}$	The distribution law is $n_i = \frac{g_i}{(e^{\alpha} e^{E_i/k_B T} - 1)}$	The distribution law is $n_i = \frac{g_i}{(e^{\alpha} e^{E_i/k_B T} + 1)}$
3. There is no restriction on the number of particles in the given state.	There is no restriction on the number of particles in a given state.	Only one particle in a given quantum state is allowed.
4. In the phase space the volume of a phase cell is not fixed.	The volume of phase cell is of the order of h^3 .	The volume of phase cell is of order of h^3 .
5. The behaviour of distribution function $f(E_i)$ against E_i is exponential.	For $E_i \gg k_B T$, exponential for $E_i \ll k_B T$, lies above M.B.	For $E_i \gg k_B T$, exponential where $E_i \gg E_F$. If $E_F \gg k_B T$ decreases abruptly near E_F .
6. The energy of an M.B. system, such as in ideal monoatomic gas, may be zero at absolute zero.	The energy may be zero at absolute zero.	The energy at absolute zero, cannot be zero because all particles cannot come down to ground state due to Pauli's exclusive principle.

NOTE

At high temperature, the Fermi energy varies slightly with temperature. In Fig. 4.9, the Fermi energy for curve at 10000 K is denoted by E'_F .